Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Convex Perturbation Paper - My Response to Your Latest Comments

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Mon, Sep 28, 2026 at 4:02 PM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Hi Ben,

Both points are in and I have verified that they are correct.
I have no more questions, so we could close this come and forth; with nothing outstanding on my side.

Best,
Ezequiel

On Sun, Sep 27, 2026 at 10:19 PM Benjamin D. Summers <bsummers@adagiollc.com> wrote:

> Hi Ezequiel,
>
> Thank you. Both points are in; revision 4 is on SSRN under the same DOI (https://dx.doi.org/10.2139/ssrn.6812461).
>
> 1. Scale-level witness. Proposition 4.10 now has two parts. Part (i) is the previous statement. Part (ii) is your construction: for divisible Γ_a and h > 0, Γ_0 = (Γ_a + hΓ_ref)/(1 + h), which is (14) at g* = −h, is a two-time family with X = −hK at every triple, D_CK(δt;Γ_0) = hM_*(δt), and D_CK(δt;Γ_g) = (1 − g)(h + g)M_*(δt) = E(g) identically, so g_first = g_term = g_1 at the scale level. The proof uses that the identity (6) is an algebraic expansion valid for every real g, and evaluates it at g = −h, where Γ_g = Γ_a. A paragraph after the proof reads (i) and (ii) as one formula with Γ_a on either side of Γ_0 along the affine line through Γ_ref, and notes that the entrywise condition is automatic for g* < 0. Remark 4.5 now cites 4.10(ii) for the attainment at the scale level and labels the λ = 0.35 configuration pointwise; the introduction, the outline, and the lead of Section 4.3 say the same. The abstract is unchanged; its claim now has the witness it needed.
> 2. Instance. With the same Glauber pair and h = 0.6: a(τ) = (e^{−2τ} − e^{−4τ})/(1 + h), M_*(δt) = 1/[16(1 + h)²] = 0.024414 for δt ≥ ln 2, D_CK(δt;Γ_0) = 0.014648, and D_CK(δt;Γ_g) = 0.024414(1 − g)(0.6 + g), which rises on [0, 0.2] before it falls (a scale-level instance of the initial rise in Remark 3.3). At ε = 0.01 the numbers come out exact: g_first = g_term = g_1 = 0.68 and g_0 = 0.744. My checks on 3,000 random triples give max ‖X + hK‖ = $3.0\cdot10^{-16}$ and max |D − E| = $6.9\cdot10^{-18}$.
> 3. Remark 6.2 now reads "for 0 < g < 1 it reduces to the fixed weight g exactly when π^Y_s = π^Z_s for all s (at g = 0 and g = 1 the weight is constant whatever the laws)."
> 4. The 1/64. You are right that it is not accidental; the manuscript now says the value is the same for every g* ∈ (0, ½], the range in which Γ_0 is stochastic.
>
> Numbering: no theorem, proposition, remark, or section number moved. The new display (16) shifts the equation numbers of Sections 5 and 6 by one, so the state-dependent class is now (19) and its bound (21); the cross-references in the manuscript are automatic.
>
> Thank you for the reviews.
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> Ben
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> On Sun, Sep 27, 2026 at 6:15 PM Ezequiel Boero <ezequiel.boero@adagioinstitute.org> wrote:
>
> > Hi Ben,